

Q1 - 25 July - Shift 1

Let $f(x) = \begin{cases} |4x^2 - 8x + 5|, & \text{if } 8x^2 - 6x + 1 \geq 0 \\ [4x^2 - 8x + 5], & \text{if } 8x^2 - 6x + 1 < 0 \end{cases}$,

where $[\alpha]$ denotes the greatest integer less than or equal to α . Then the number of points in $\mathbb{R}$ where f is not differentiable is _____.

Space for your notes:

Q2 - 26 July - Shift 1

If $f(x) = \begin{cases} x + a, & x \leq 0 \\ |x - 4|, & x > 0 \end{cases}$ and

$g(x) = \begin{cases} x + 1, & x < 0 \\ (x - 4)^2 + b, & x \geq 0 \end{cases}$

are continuous on $\mathbb{R}$, then $(g \circ f)(2) + (f \circ g)(-2)$ is equal to :

- (A) -10 (B) 10
(C) 8 (D) -8

Space for your notes:

Q3 - 27 July - Shift 2

If for $p \neq q \neq 0$, then function

$f(x) = \frac{\sqrt[7]{p(729 + x)} - 3}{\sqrt[3]{729 + qx} - 9}$ is continuous at $x = 0$, then:

- (A) $7pq f(0) - 1 = 0$ (B) $63q f(0) - p^2 = 0$
(C) $21q f(0) - p^2 = 0$ (D) $7pq f(0) - 9 = 0$

Space for your notes:

Q4 - 28 July - Shift 2

The function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \lim_{n \rightarrow \infty} \frac{\cos(2\pi x) - x^{2n} \sin(x-1)}{1 + x^{2n+1} - x^{2n}}$$

is continuous for all x in

- (A) $\mathbb{R} - \{-1\}$ (B) $\mathbb{R} - \{-1, 1\}$
(C) $\mathbb{R} - \{1\}$ (D) $\mathbb{R} - \{0\}$

Space for your notes:

Q5 - 29 July - Shift 1

The number of points, where the function

$f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = |x-1| \cos |x-2| \sin |x-1| + (x-3)|x^2-5x+4|$, is **NOT** differentiable, is :

- (A) 1 (B) 2 (C) 3 (D) 4

Space for your notes:

Q6 - 29 July - Shift 2

Let the function

$$f(x) = \begin{cases} \frac{\log_e(1+5x) - \log_e(1+\alpha x)}{x} & ; \text{if } x \neq 0 \\ 10 & ; \text{if } x = 0 \end{cases}$$

be continuous at $x = 0$.

The α is equal to :

- (A) 10 (B) -10
(C) 5 (D) -5

Space for your notes:

Q7 - 29 July - Shift 2

Questions

MathonGo

If $[t]$ denotes the greatest integer $\leq t$, then number of points, at which the function $f(x) = 4|2x + 3| + 9\left[x + \frac{1}{2}\right] - 12[x + 20]$ is not differentiable in the open interval $(-20, 20)$, is ____.

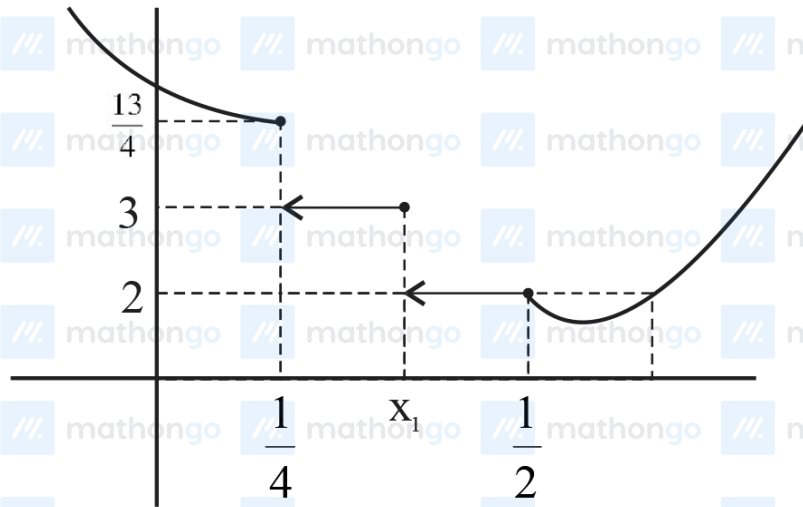
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Answer Key**Q1 (3)****Q2 (D)****Q3 (B)****Q4 (B)****Q5 (B)****Q6 (D)****Q7 (79)**

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Q1 (3)



ND at $\frac{1}{4}, x_1, \frac{1}{2}$

Q2 (D)

$$f(x) = \begin{cases} x + a & ; x \leq 0 \\ |x - 4| & ; x > 0 \end{cases} ; g(x) = \begin{cases} x + 1 & ; x < 0 \\ (x - 4)^2 + b & ; x \geq 0 \end{cases}$$

For continuity $a = 4$ and $b = -15$

$$g(f(2)) + f(g(-2))$$

$$= g(2) + f(-1) = -8$$

Q3 (B)

$$f(0) = \lim_{x \rightarrow 0} f(x)$$

Limit should be $\frac{0}{0}$ form

$$\text{So, } \sqrt[7]{p \cdot 729} - 3 = 0 \Rightarrow p \cdot 3^6 = 3^7 \Rightarrow p = 3$$

$$\text{Now, } f(0) = \lim_{x \rightarrow 0} \frac{\sqrt[7]{3(3^6 + x)} - 3}{\sqrt[3]{3^6 + qx} - 9}$$

$$= \lim_{x \rightarrow 0} \frac{3 \left[\left(1 + \frac{x}{3^6} \right)^{1/7} - 1 \right]}{9 \left[\left(1 + \frac{qx}{3^6} \right)^{1/3} - 1 \right]} = \frac{3}{9} \times \frac{\frac{1}{7 \cdot 3^6}}{\frac{q}{3 \cdot 3^6}}$$

$$\Rightarrow f(0) = \frac{1}{3} \times \frac{3}{7q} = \frac{1}{7q}$$

$$\Rightarrow 7qf(0) - 1 = 0$$

$$\Rightarrow 7 \cdot p^2 \cdot qf(0) - p^2 = 0 \text{ (for option)}$$

$$\Rightarrow 63qf(0) - p^2 = 0$$

Q4 (B)

n should be given as a natural number.

$$f(x) = \begin{cases} \frac{-\sin(x-1)}{x-1} & x < -1 \\ -(\sin 2 + 1) & x = -1 \\ \cos 2\pi x & -1 < x < 1 \\ 1 & x = 1 \\ \frac{-\sin(x-1)}{x-1} & x > 1 \end{cases}$$

$f(x)$ is discontinuous at $x = -1$ and $x = 1$

Q5 (B)

$$f(x) = |x-1| \cos |x-2| \sin |x-1| + (x-3)|x^2-5x$$

$$+ 4|$$

$$= |x-1| \cos |x-2| \sin |x-1| + (x-3)|x-1||x-4|$$

$$= |x-1| [\cos |x-2| \sin |x-1| + (x-3)|x-4|]$$

Non differentiable at $x = 1$ and $x = 4$.

Q6 (D)

$$f(x) = \begin{cases} \frac{\ln(1+5x) - \ln(1+\alpha x)}{x} & ; x \neq 0 \\ 10 & ; x = 0 \end{cases}$$

$$\lim_{x \rightarrow 0} \frac{\ln(1+5x) - \ln(1+\alpha x)}{x} = 10$$

Using expansion

$$\lim_{x \rightarrow 0} \frac{(5x + \dots) - (\alpha x + \dots)}{x} = 10$$

$$5 - \alpha = 10 \Rightarrow \alpha = -5$$

Q7 (79)

$$f(x) = 4|2x + 3| + 9 \left[x + \frac{1}{2} \right] - 12[x + 20]$$

$$x \in (-20, 20)$$

$f(x)$ is not Diff. at $x = I \in \{-19, -18, \dots, 0, \dots, 19\} = 39$

at $x = I + \frac{1}{2}$, $f(x)$ Non diff. at 39 points

Check at $x = \frac{-3}{2}$ Discount at $x = \frac{-3}{2} \therefore N, R(1)$

No. of point of non-differentiability

$$= 39 + 39 + 1 = 79$$